THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 13: **Algebra Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Think of a number n. Add 3, multiply by 5, then subtract 15. The result is: 1

(a) $5n$ **(b)** $5n + 3$ **(c)** $3n$ **(d)** $n + 3$

2. The sum of 3 consecutive integers starting with n is: 1

(a) $3n + 4$ **(b)** $3n$ **(c)** $n + 3$ **(d)** $3n + 3$

3. In a number pyramid each block is the sum of the two below it. The bottom row is 1, 2, 3, 4. The top block is: 1

(a) 22 **(b)** 10 **(c)** 15 **(d)** 20

4. In a calendar the nine dates in a 3×3 block have the middle date 10. Their sum is: 1

(a) 99 **(b)** 90 **(c)** 81 **(d)** 30

5. The value of $8 \times 8 - 7 \times 9$ is: 1

(a) 8 **(b)** 2 **(c)** 1 **(d)** 0

6. For any n, $(n + 1)^2 - (n - 1)^2$ equals: 1

(a) 2 **(b)** 4 **(c)** $4n$ **(d)** $2n$

7. Think of a number n. Add 6, multiply by 4, then subtract 24. The result is: 1

(a) $4n + 6$ **(b)** $n + 6$ **(c)** $6n$ **(d)** $4n$

8. The sum of 4 consecutive integers starting with n is: 1

(a) $4n$ **(b)** $4n + 6$ **(c)** $n + 6$ **(d)** $4n + 7$

9. In a number pyramid each block is the sum of the two below it. The bottom row is 2, 5, 1, 3. The top block is: 1

(a) 11 **(b)** 17 **(c)** 22 **(d)** 23

10. In a calendar the nine dates in a 3×3 block have the middle date 12. Their sum is: 1

(a) 117 **(b)** 108 **(c)** 36 **(d)** 99

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Show that the sum of 16 and its reverse is divisible by 11. 2

12. Think of a number n, add 4, multiply by 3, subtract 12, divide by 3. Show that the answer is n. 2

13. Express the sum of two consecutive odd numbers starting with $2n + 1$ and show that it is a multiple of 4. 2

14. Show that the sum of 38 and its reverse is divisible by 11. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a number pyramid with bottom row 2, 3, 1, 4, each block is the sum of the two below it. Find all the blocks. 3

16. Show that the difference of 83 and its reverse 38 is a multiple of 9. 3

17. A 3×3 block of calendar dates has middle date 11. Write the nine dates and show that their sum is 9 times the middle date. 3

OR

In a number pyramid with bottom row 5, 2, 3, 1, each block is the sum of the two below it. Find all the blocks. 3

18. Show that the difference of 94 and its reverse 49 is a multiple of 9. 3

OR

A 3×3 block of calendar dates has middle date 13. Write the nine dates and show that their sum is 9 times the middle date. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Show that the sum of 3 consecutive integers starting with n is $3n + 3$ and verify it for $n = 14$. 5

OR

Think of a number n. Double it, add 10, halve the result and subtract n. Show algebraically that the final answer is always 5. Verify for $n = 7$. 5

20. Show that the sum of 4 consecutive integers starting with n is $4n + 6$ and verify it for $n = 9$. 5

OR

Think of a number n. Double it, add 14, halve the result and subtract n. Show algebraically that the final answer is always 7. Verify for $n = 9$. 5

END OF QUESTION PAPER — Total: 40 Marks

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